

Scientific Reading – Guiding Questions

As part of your first milestone, you will need to review a computational method from a scientific paper or a professional blog post and explain how it, or an adapted version of it, can be applied to our project's specific challenges. There is a list of suggestions in Moodle ((מאמרים ופוסטים עם רעיונות לפרוייקט (ולמייילסטון 1)), you can either use one of them or find your own interesting idea.

The sources you choose will vary. You may read papers specifically designing pipelines for microbiome data, while you can also explore general-purpose machine learning methods. This guide is designed to help you extract the most important information regardless of the source material.

When working with different types of sources, it is important to keep in mind that blog posts and scientific papers are written with different goals and styles, and should therefore be approached slightly differently. Blog posts are often designed to build intuition and are typically more accessible, offering clear explanations and practical insights. However, they may simplify certain aspects, omit technical details, or ignore edge cases, so it is important to think critically about what might be missing or oversimplified. In contrast, scientific papers tend to be more formal and comprehensive, providing a complete and rigorous description of the method. This can make them more challenging to read, but they also contain the full details needed for a deeper understanding. When reading a paper, it is especially helpful to focus on figures, method diagrams, and experimental results, as these often convey the core ideas more clearly than the text alone.

Keep in mind: You do not need to understand every single mathematical derivation or complex equation. Your goal is to grasp the core intuition behind the method, understand how the authors validated it, and clearly define how it will inspire the pipeline you build. Generative AI tools such as ChatGPT, Gemini, and NotebookLM can be very helpful when reading and understanding papers or blog posts. You are encouraged to use them to clarify concepts, summarize sections, or get alternative explanations. However, these tools should be used to support your understanding, not replace it. Make sure that you still read the original source and develop your own understanding of the method, as this will be essential for applying it correctly in your project.

For your Milestone 1 presentation, clearly communicate both your understanding of the method and how it connects to your project by a concise overview and an intuitive explanation of how the method works (preferably as a simple pipeline). Summarize the key results, discuss the main strengths and limitations, and explain how you plan to use or adapt the method in your pipeline. Keep the presentation short and focused, just a few clear slides that can be presented in 10–20 minutes, prioritizing the most relevant ideas

rather than full coverage. Aim for clarity over completeness: if you cannot explain the method simply, you likely do not understand it well enough. Using diagrams or visualizations is strongly encouraged to convey ideas effectively.

The purpose of the following guiding questions is to direct your reading and prepare for your presentation, you do not have to use them, but you may find them useful to organize your reading and train of thought. Not all questions will perfectly fit every paper or blog post, so focus on the ones that best apply to your chosen source.

If you have any questions, regarding the paper selection, the methods, the presentation or any other topic – feel free to contact us.

The Big Picture: Motivation and Problem Definition

Before diving into the mechanics, understand why this method exists.

- What is the primary problem or bottleneck the authors are trying to solve?
- Why is this problem important?
- Why are traditional, simpler, or "naïve" approaches insufficient for this problem?
- Does the source address missing data, multi-modal or diagonal integration, or distance estimation? How does their specific problem relate to our challenge of dealing with missing omics modalities?

The Core Method: How It Works

Focus on the intuition and the architecture of the solution.

- What is the high-level concept or algorithmic approach (Try to explain it in 2-3 simple sentences)? What makes it different from simpler or standard approaches?
- What are the required inputs for this method, and what is the exact output? What are the main steps in the pipeline?
- Identify the building blocks: data representation (features, embeddings, graphs, etc.), model type (e.g., regression, neural network, clustering, etc.), Objective function, Optimization/training process. What are the most important components? Which parts are essential vs. optional?
- What assumptions does the model make about the data? (For example, does it assume data is normally distributed, or that samples are independent?) When might this method fail?

- If you had to draw a flowchart of this method, what would the 3 to 5 main steps be?

Validation: Proving It Works

A method is only as good as its validation. Look at how the authors proved their approach is effective.

- What datasets did they use to test their method? Were they biological datasets, or synthetic/general data?
- What evaluation metrics did they use to define "success"?
- What baselines did they compare their method against?
- What does this method do well? When is it especially useful?
- What were the limitations or weaknesses acknowledged by the authors?

Adaptation: Connecting It to Our Project

This is the most crucial part for your milestone. You should bridge the gap between their work and your pipeline.

- Why is this method relevant to your problem?
- Which part of your pipeline could it improve (Handling missing data? Learning representations? Computing distances? Integrating modalities?)?
- How does this method inspire your project? Do you plan to Use the method as-is? Modify it? Combine it with other approaches? Have a completely different approach?
- What changes are needed to apply it?
- At what stage of your pipeline will this method be implemented (e.g., preprocessing, imputation, dimensionality reduction, or distance calculation)?
- How will you implement it? What inputs will you use? How will you evaluate it?

Practical Implementation

Consider the engineering reality of using this method.

- Did the authors provide accessible, open-source code (e.g., a GitHub repository or Python package)?

- Does the method seem computationally heavy? Will it run efficiently on the scale of our dataset within a reasonable timeframe?
- What Python libraries or tools will you need to learn or use to implement this idea?